

Real Time Helmet Violation Detection Using Deep Learning

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Agenda

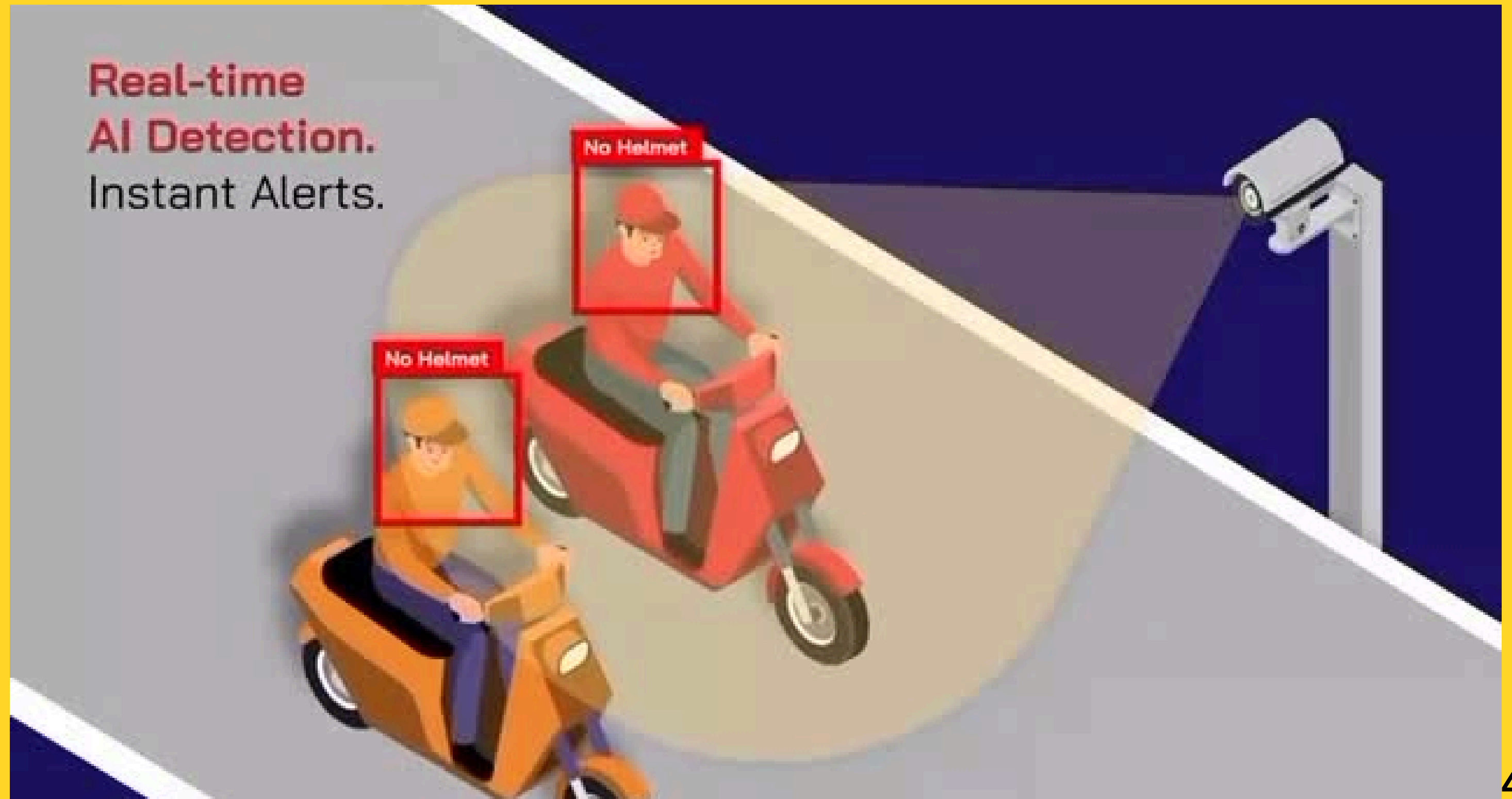
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DOMAIN:

ARTIFICIAL INTELLIGENCE (AI) / COMPUTER VISION / DEEP LEARNING

This project falls under the domain of **Artificial Intelligence**, specifically focusing on Computer Vision and Deep Learning. It uses AI techniques to automatically analyze video footage, detect motorcycles, and identify helmet usage in real time, making it a practical application of intelligent image and video analysis for traffic monitoring and road safety enforcement.

SHORT VIDEO:



ABSTRACT

Artificial Intelligence (AI) is increasingly being applied to solve everyday societal challenges by enabling machines to perform tasks that require human-like intelligence. Within this domain, deep learning has emerged as a powerful technique in computer vision, allowing systems to automatically recognize, classify, and analyze visual data for improved decision-making.

In the existing system, researchers proposed an artificial intelligence multitier framework with a lightweight classifier for helmetless biker detection. Their method follows a two-stage process: first detecting motorbike riders from traffic footage, and then classifying whether they are wearing helmets. By designing a lightweight model, the system reduces computational cost, enabling faster detection with acceptable accuracy. The authors also built custom datasets to train and evaluate the system under different conditions. While this approach achieved promising results, it primarily addressed only helmet detection and was tested under limited scenarios.

The disadvantage of the existing work is that it does not consider broader violations such as triple riding, mobile phone usage, or varying environmental conditions like night and rain. Our proposed work addresses these gaps by utilizing detectnet and YOLO v8 algorithm, while also incorporating robust datasets to cover diverse scenarios. Additionally, the system will generate automated alerts and send them to authorized institutions or organizations, triggering warning messages to riders. This ensures higher accuracy, real-time enforcement, and practical deployment for improving road safety.

Keywords: Artificial Intelligence, Deep Learning, Helmet Detection, Traffic Safety, Computer Vision.

Objective:

WHY?

To **reduce road accidents and fatalities** caused by riders not wearing helmets and to encourage compliance with traffic safety rules.

HOW?

By using a camera along with computer vision techniques and deep learning models, the system monitors riders, detects violations, and triggers an immediate beep alert as a warning.

WHOM GETS BENEFITS:

Riders benefit by receiving instant safety alerts, society benefits from reduced accident risks, and authorities benefit from an intelligent tool to promote safe riding practices.

Introduction

Road accidents involving two-wheeler riders are often caused by not wearing helmets. Manual checking of helmet violations is inefficient and time-consuming. This project introduces a **real-time helmet detection system** using **deep learning** to automatically identify riders without helmets from live video feeds. The system detects motorcycles and classifies helmet usage accurately, helping traffic authorities enforce rules and **improve road safety efficiently**.

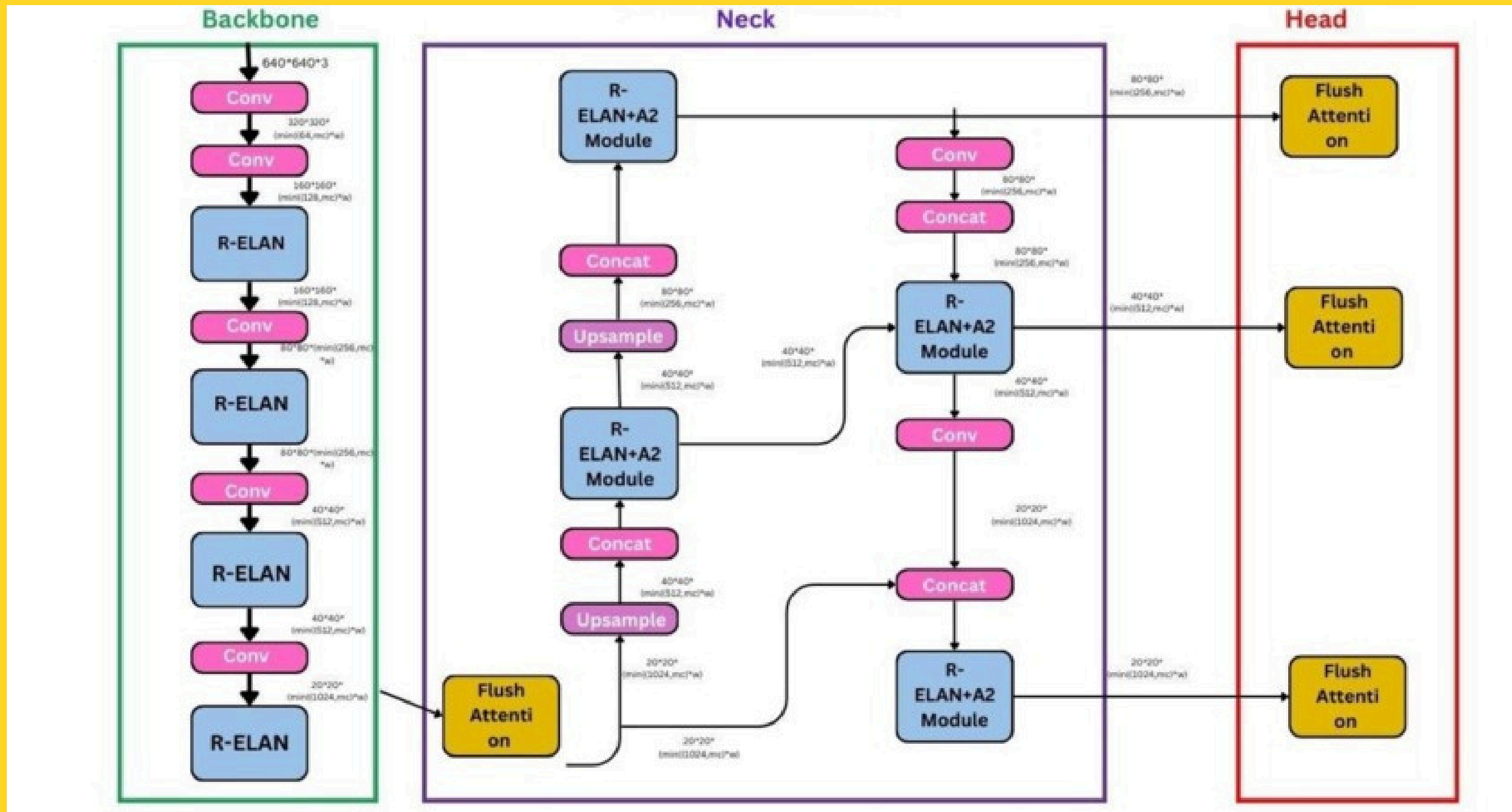




Existing System

- In the existing system, helmet violation detection is mostly done manually by traffic police or through basic image processing techniques. **Traffic officers monitor CCTV footage** or roadside cameras and identify violations visually, which is time-consuming, less accurate, and prone to human error.
- Some traditional systems use simple color or shape-based image processing, but they fail under poor lighting, different helmet styles, or complex backgrounds.
- Overall, **existing systems Lack automation**, real-time processing, and accuracy, highlighting the need for an advanced deep learning-based approach to detect helmet violations more efficiently.

ARCHITECTURE DIAGRAM:



ADVANTAGES

- Superior Real-Time Accuracy (Speed & Precision)
- Enhanced Small Object Detection
- Robustness to Visual Challenges
- Efficient Model Deployment (Optimization)

DISADVANTAGES

- High Computational Cost & Hardware Dependency
- Data Requirement for New Features
- Privacy and Ethical Concerns
- False Positives from Small Object Confusion

PROBLEM STATEMENT:

- Motorcycle accidents are increasing mainly due to riders not wearing helmets or using unsafe helmets, leading to severe head injuries and fatalities.
- Manual monitoring by traffic police to detect helmetless riders is inefficient, time-consuming, and prone to human error.
- Most current systems only detect the presence or absence of helmets but cannot assess helmet quality (safe, unsafe, or inappropriate).
- There is a need for a lightweight, accurate, and automated AI-based system capable of detecting and classifying helmetless riders and helmet quality in real-time to enhance road safety and law enforcement efficiency.

LITERATURE SURVEY:

TITLE : Automated detection of bicycle helmets using deep learning

AUTHORS: Siebert F.W., Riis C., Janstrup K.H., Lin H., et al.

YEAR: 2024

DESCRIPTION:

This study presents a computer vision approach to automatically detect bicycle helmets from roadside camera footage in Copenhagen. A dataset of ~4,000 cyclists, supplemented with web images, was annotated for active cyclist detection and helmet use. The authors trained and tested deep learning models to estimate helmet-wearing rates, comparing results against manual human observations. The system demonstrated near-human accuracy when tested on the same site, with a minimal underestimation of 0.52%. However, accuracy dropped when applying models trained at one site to another, highlighting the challenges of cross-site generalization in real-world deployments.

CONCLUSION:

Automated helmet-use detection is feasible and highly accurate under controlled conditions. However, cross-site transfer remains a limitation, and improving robustness across diverse environments is crucial for large-scale deployment.

TITLE: Effect of helmet use on maxillofacial injuries: a systematic review and meta-analysis

AUTHORS: Stassen H.S., Atalik T., Haagsma J.A., Wolvius E.B., et al.

YEAR: 2024

DESCRIPTION:

This paper systematically reviews and meta-analyzes studies examining the relationship between helmet use and maxillofacial injuries in bicycle and scooter crashes. Fourteen studies were included, of which eleven contributed to the quantitative synthesis. The pooled odds ratio ($OR \approx 0.68$) indicates that helmets significantly reduce the risk of maxillofacial trauma. The included studies were predominantly observational, with variable quality ratings. The review emphasizes that while evidence supports protective effects, research is limited by heterogeneity, older datasets, and lack of focus on e-bikes and e-scooters, which have unique injury patterns and growing urban prevalence.

CONCLUSION:

Helmets provide significant protection against maxillofacial injuries. However, higher-quality and more recent studies, particularly addressing e-bikes and e-scooters, are needed to strengthen evidence and guide policy.

TITLE: AI-Based Helmet Violation Detection for Traffic Management System

AUTHORS: Said Y., Alassaf Y., Ghodhbani R., Alsariera Y.A., et al.

YEAR: 2024

DESCRIPTION:

This work introduces an AI-driven helmet violation detection framework aimed at improving road safety and assisting traffic enforcement authorities. The system is built on a modified PerspectiveNet model with EfficientNet v2 as the backbone, balancing detection accuracy and computational efficiency for real-time surveillance. It is evaluated using public traffic datasets and demonstrates a 95.2% detection accuracy, with reduced model complexity compared to conventional deep learning approaches. The authors highlight scalability in dense urban traffic, capability for real-time operation, and potential for integration into broader intelligent transportation systems.

CONCLUSION:

The study validates the feasibility of using lightweight deep learning models for helmet violation detection in real-world traffic conditions. It emphasizes scalability and efficiency, making the system suitable for real-time deployment in smart cities.

TITLE: DetectNet_v2 + YOLOv8 for helmet violation detection and license plate extraction

AUTHORS: Deshpande U.U., Goh G.K.O.M., Araujo S.D.C.S., et al.

YEAR: 2025

DESCRIPTION:

This paper proposes a two-stage helmet violation detection pipeline tailored for Indian traffic conditions. In the first stage, DetectNet_v2 with a ResNet18 backbone detects motorcycle riders, while the second stage employs YOLOv8 to classify helmet use and detect license plates. An OCR module extracts license numbers for enforcement. A custom dataset of ~1,715 images captured in varied lighting and traffic conditions is used for training and evaluation. The system achieves impressive accuracy: helmet detection (98.56%) and license plate recognition (97.6%), with competitive inference times for near real-time application in urban surveillance systems.

CONCLUSION:

The two-stage pipeline demonstrates robust performance for helmet violation detection and license plate recognition. Its strong accuracy and practical orientation highlight its suitability for deployment in Indian smart-city surveillance networks.

TITLE: Smart Road Surveillance: Helmet Detection and Triple Ride Detection Using CNN and YOLO

AUTHORS: Ranganayaki J., Poojitha V., Dikshith Y.

YEAR: 2025

DESCRIPTION:

This engineering-oriented paper presents a smart road surveillance system combining YOLO and CNN for detecting helmet violations and triple-riding cases on motorcycles. The system includes license plate recognition using Tesseract OCR, enabling automatic identification of violators. The implementation is tailored for resource-constrained environments, focusing on practical deployment in developing regions. The authors describe the system architecture, dataset preparation, and provide screenshots of real-world tests. While performance metrics are not as comprehensive as in research-focused works, the emphasis is on integrating multiple violation detection features into a functional prototype system.

CONCLUSION:

The proposed system demonstrates the integration of helmet and triple-ride detection into a single surveillance framework. Its strength lies in practical feasibility and low-cost deployment, though further validation with larger datasets is needed.

LITERATURE SURVEY TABLE:

TITLE	TECHNIQUE USED	CONCLUSION
AI-Based Helmet Violation Detection for Traffic Management System	Optimized PerspectiveNet with EfficientNet v2 backbone, deep learning-based real-time helmet violation detection.	We conclude that architecture optimization enables accurate real-time traffic monitoring, guiding our project toward efficient and scalable safety applications.
Computer-vision based automatic rider helmet violation detection and vehicle identification in Indian smart city scenarios using NVIDIA TAO toolkit and YOLOv8	Two-step approach: DetectNet_v2 (ResNet18 + TAO) for rider detection, YOLOv8 for helmet + number plate detection, OCR for plate recognition; Custom dataset of 1,715 images	We conclude that combining multi-model deep learning improves reliability. This supports our project vision to use a multi-model approach for accurate detection, even in complex real-time traffic scenarios.

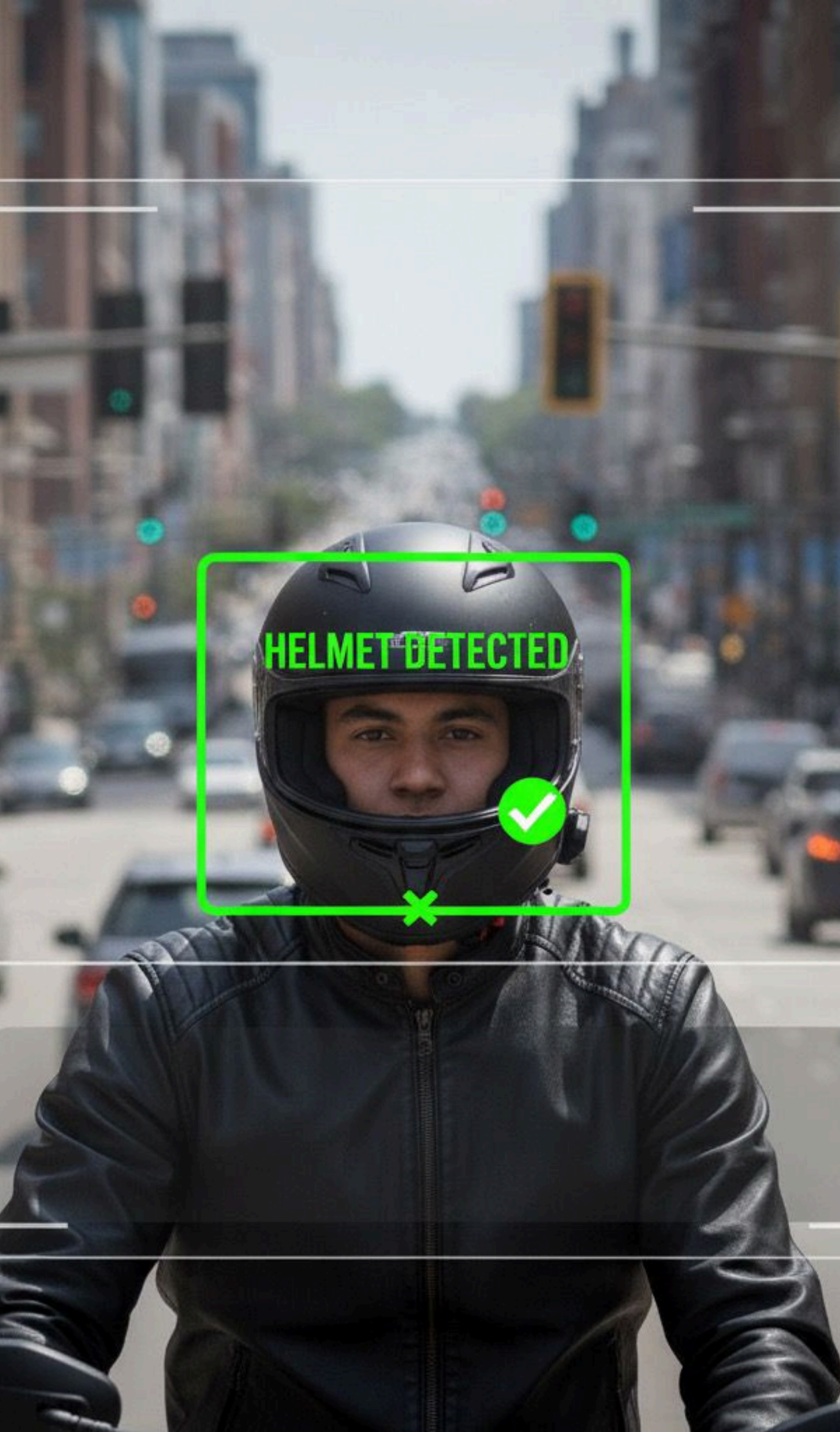
TITLE	TECHNIQUE USED	CONCLUSION
Automated detection of bicycle helmets using deep learning	Computer vision + state-of-the-art object detection algorithms trained on annotated cyclist datasets, real-world traffic video analysis.	We conclude that helmet detection can scale beyond motorbikes to bicycles, proving our project can extend to multiple road user categories for broader safety enforcement.
Smart Road Surveillance: Helmet Detection and Triple Ride Detection Using CNN and YOLO	YOLOv5 for motorcycle, rider, and plate detection; CNN for helmet/no-helmet classification; OCR (Tesseract) for license plate recognition; real-time triple ride detection	We conclude that integrating multiple detection tasks (helmet, license plate, triple ride) into a unified system enhances road safety enforcement.
An AI multitier system with lightweight classifier for automated helmetless biker detection	Custom CNN model BikeNet-12, datasets with helmet/no-helmet/safe-unsafe helmet classes, YOLOv8 for object detection in varied conditions.	We conclude that lightweight models can deliver high accuracy with less computation, highlighting efficiency as a key factor for our system's real-world deployment.

Problem Statement:

- Current AI-based systems use lightweight classifiers in a two-stage process — first detecting motorbike riders, then classifying helmet usage.
- These methods focus only on helmet detection, often under limited conditions.
- They do not address other traffic violations such as:
 1. Triple riding
 2. Mobile phone usage by riders
- Existing models struggle under challenging conditions like:
 1. Night-time
 2. Rain or poor lighting
- They also lack automated enforcement features, meaning there is no instant alert to authorities or warning to offenders.

Current Status

- Accurately identify motorcyclists and differentiate between riders with and without helmets in live traffic videos, even under variable environmental conditions.
- Minimize false positives and negatives by leveraging advanced object detection and optimized training with large, diverse datasets.
- Enable seamless integration with modules for license plate recognition and automated alert generation, making real-time enforcement feasible for smart city applications



CONCLUSION:

- Significant Leap in Performance: YOLOv12 represents a significant leap in real-time helmet violation detection, blending state-of-the-art object detection accuracy with faster inference speeds.
- Robustness: It is robust in handling diverse environmental conditions and complex traffic scenes, ensuring reliable helmet detection and rider identification.
- Powerful Tool for Road Safety: YOLOv12 stands out as a powerful tool for enhancing road safety, reducing helmet-related traffic violations, and supporting smart city traffic management efforts.

REFERENCE PAPER:

01. Yahia Said, Yahya Alassaf, Refka Ghodhbani, Yazan Ahmad Alsariera.
"AI-Based Helmet Violation Detection for Traffic Management System"
02. J. Ranganayaki, Vunnam Poojitha, and Yerroju Dikshith
"Smart Road Surveillance: Helmet Detection and Triple Ride Detection Using CNN and YOLO"
03. Uttam U. Deshpande, Goh Kah Ong Michael, Sufola Das Chagas Silva Araujo.
"Computer-vision based automatic rider helmet violation detection and vehicle identification in Indian smart city scenarios using NVIDIA TAO toolkit and YOLOv8"
04. J.Ranganayaki, Vunnam Poojitha, and Yerroju Dikshith
"Helmet Detection and Triple Ride Detection Using CNN and YOLO"

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Thank You for Attending!